Lecture 3
Binary Images and Morphological Image Processing

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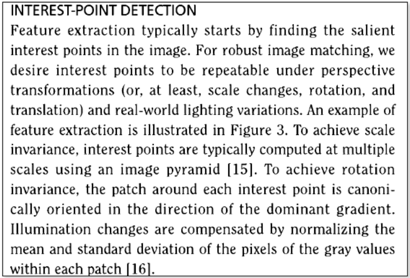
one sample
max/min in image enhancement

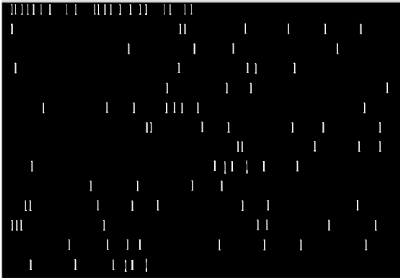
An example of recognition using opening

Binary image f



INTEREST-POINT DETECTION
Feature extraction typically starts by finding the salient interest points in the image. For robust image matching, we desire interest points to be repeatable under perspective transformations (or, at least, scale changes, rotation, and translation) and real-world lighting variations. An example of feature extraction is illustrated in Figure 3. To achieve scale invariance, interest points are typically computed at multiple scales using an image pyramid [15]. To achieve rotation invariance, the patch around each interest point is canonically oriented in the direction of the dominant gradient. Illumination changes are compensated by normalizing the mean and standard deviation of the pixels of the gray values within each patch [16].

$$open\left(NOT\left[f\right],W\right) = dilate\left(erode\left(NOT\left[f\right],W\right),W\right)$$

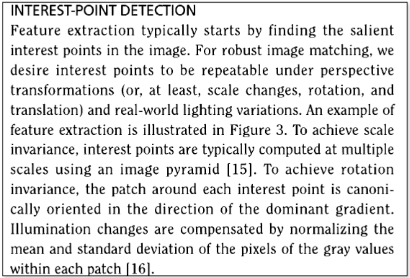


Structuring element W

Hit-Miss Filter

Binary image f

$$dilate\left(erode\left(NOT\left[f\right],V\right)\&\,erode\left(f,W\right),W\right)$$



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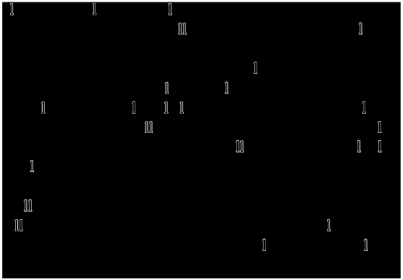
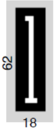


Structuring element V



Structuring element W

Boundary Extraction

1. $\beta(A)$ denotes the boundary of a set A.
2. The boundary is extracted by using

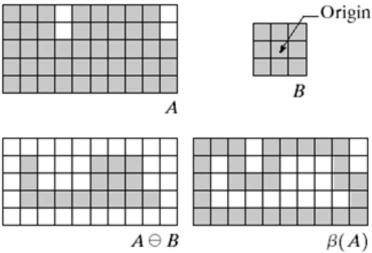
$$\beta(A) = A - (A \ominus B).$$

3. The boundary is the difference between the object and the "eroded" object.

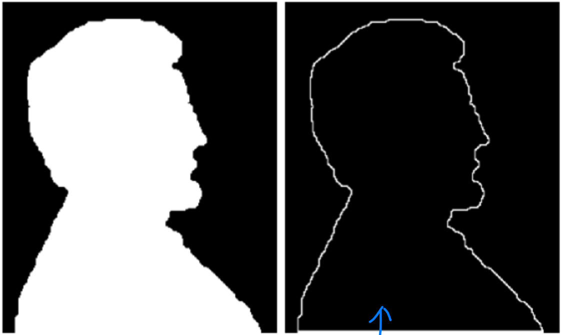
Shaded region $= 1$, white region $= 0$

a b
c d

FIGURE 9.13 (a) Set A. (b) Structuring element B. (c) A eroded by B. (d) Boundary, given by the set difference between A and its erosion.



edge detector



a b

FIGURE 9.14 (a) A simple binary image, with 1's represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

grow it within boundary

Region Filling

1. Beginning with a point p inside the boundary, the objective is to fill the entire region with 1s.
2. The algorithm is defined as follows.

$$X_k = (X_{k-1} \oplus B) \cap A^c.$$

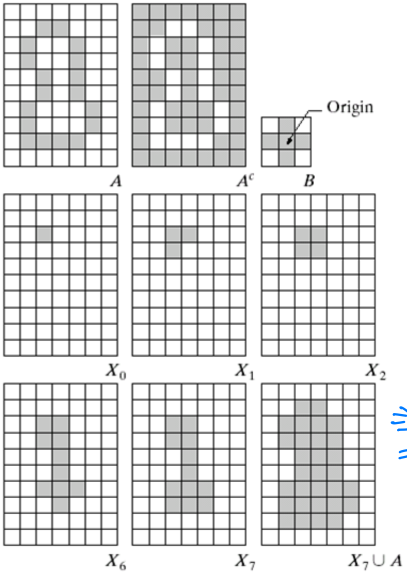
3. k=1,2,3,...
4. $X_0 = p$
5. B is the symmetric structuring element.
6. Intersection with A^c constrains the result to be inside the region of interest.
7. The algorithm stops when $X_k = X_{k-1}$. This means the algorithm stops when there is no change in the region size.

Shaded region = 1, white region = 0

reconstruct!

a b c
d e f
g h i

FIGURE 9.15
Region filling.
(a) Set A.
(b) Complement of A.
(c) Structuring element B.
(d) Initial point inside the boundary.
(e)–(h) Various steps of Eq. (9.5-2).
(i) Final result [union of (a) and (h)].



$\Rightarrow X_8 \cup A$
$\Rightarrow$ stop

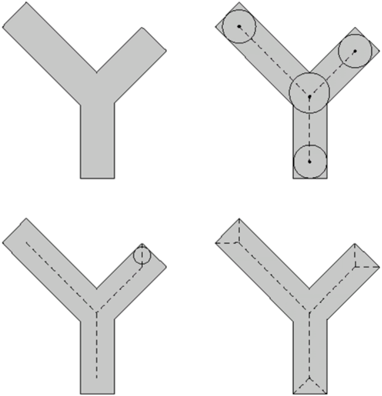
a b c

FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the region-filling algorithm). (b) Result of filling that region (c) Result of filling all regions.

Skeletons I

1. Definition of a skeleton, $S(A)$, of a set A is as follows
   - If
     - z is a point of $S(A)$ and
     - $(D)_z$ is the largest disk centered at z and contained in A,
   - then
     - one cannot find a larger disk (not necessarily centered at z) containing $(D)_z$ and included in A, and
     - the disk $(D)_z$ is called *maximum disk*.
     - the disk $(D)_z$ touches the boundary of A at two or more different places.

Skeletons II



a b
c d

FIGURE 9.23
(a) Set A.
(b) Various positions of maximum disks with centers on the skeleton of A.
(c) Another maximum disk on a different segment of the skeleton of A.
(d) Complete skeleton.

Skeletons III

2. The algorithm is given by

$$S(A) = \bigcup_{k=0}^{K} S_k(A).$$

with

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B.$$

where B is a structuring element, and

$$(A \ominus kB) = (\ldots((A \ominus B) \ominus B)\ldots) \ominus B.$$

k successive erosions of A by B.

3. K is the last iterative step before A erodes to an empty set.

$$K = \max\{k | (A \ominus kB) \neq \emptyset\}.$$

Reconstruction from skeletons

1. A can be reconstructed by using the equation,

$$A = \bigcup_{k=0}^{K} (S_k(A) \oplus kB),$$

union!

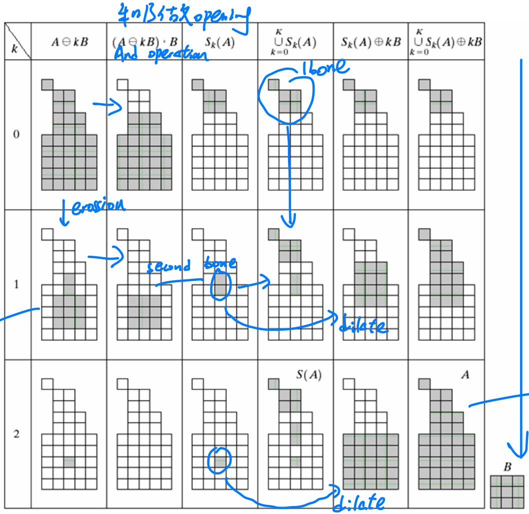
dilation for each version of skeleton

where

$$(S_k(A) \oplus kB) = (\ldots((S_k(A) \oplus B) \oplus B) \oplus \ldots) \oplus B,$$

k successive dilations of $S_k(A)$.

new image!



total 3 bones

reconstruct!

FIGURE 9.24 Implementation of Eqs. (9.5-11) through (9.5-15). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.

Gray-Scale Images

bright and dark!

- In gray scale images on the contrary to binary images we deal with digital image functions of the form f(x,y) as an input image and b(x,y) as a structuring element.
- (x,y) are integers from Z*Z that represent a coordinates in the image.
- f(x,y) and b(x,y) are functions that assign gray level value to each distinct pair of coordinates.
- For example the domain of gray values can be 0-255, whereas 0 is black, 255- is white.

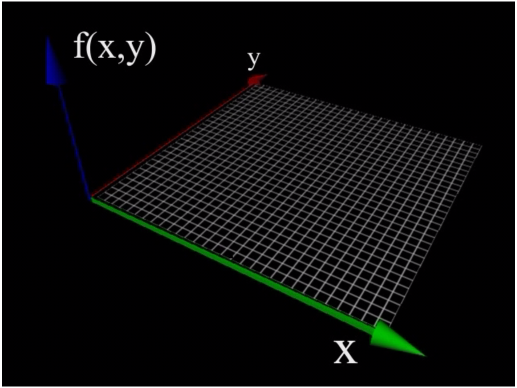
(no energy)

exactly energy!

digitized version

8 bits

Gray-Scale Images



Dilation/erosion for gray-level images

look brighter

- Explicit decomposition into threshold sets not required in practice
- Flat dilation operator: local maximum over window W manipulate

$$g[x,y] = \max\left\{W^{(-)}\left\{f[x,y]\right\}\right\} := dilate(f,W)$$

- Flat erosion operator: local minimum over window W

$$g[x,y] = \min\left\{W^{(+)}\left\{f[x,y]\right\}\right\} := erode(f,\boxed{W})$$

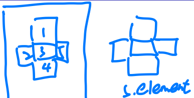
shape changed ↳ patch

- Binary dilation/erosion operators contained as special case

pass into original all pixels
minimum
= reduce lightness of whole image

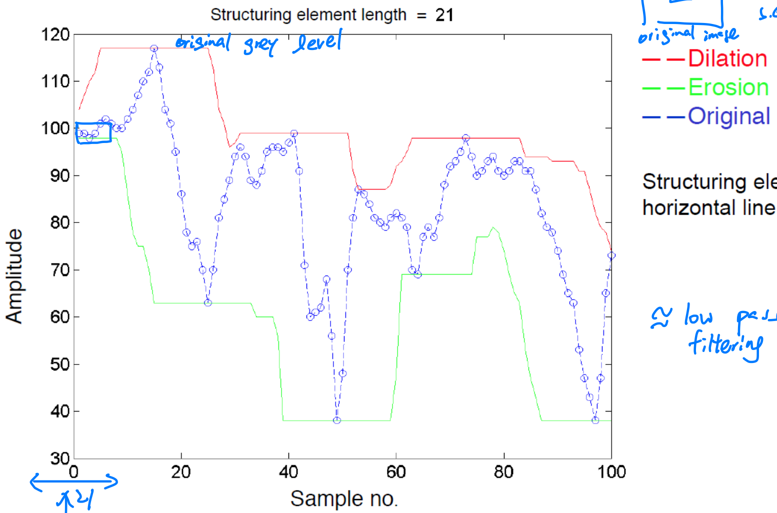
1-d illustration of erosion and dilation

one scan light



original image

s.element



Structuring element: horizontal line

≈ low pass filtering

21

check $\frac{21}{2}$ ≈ 10 neighbours

Image example

blur, ignore difference, brighter

Original

Dilation

within

Erosion

Flat dilation with different structuring elements

dilation and erosion
is irreversible

Original


Diamond


Disk


20 degree line

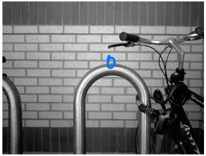

9 points


2 horizontal lines

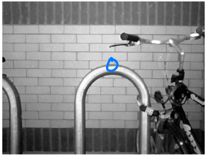
Dilation ⇒ bright filter
erosion ⇒ dark filter

Edge detector in gray level image

brighter



original f



dilation g

g-f

contour, highlight all details!

(g-f) thresholded

focus on major objects

≈ contrast

Dilation vs. Erosion for Gray-Scale Images

- The general effects of performing dilation on a gray scale image is twofold:
  - If all the values of the structuring elements are positive, the output image tends to be BRIGHTER than the input.
  - DARK details either are reduced or elimanted, depending on how their values and shape relate to the structuring element used for dilation
- The general effects of performing erosion on a gray scale image:
  - If all elements of the structuring element are positive, the output image tends to be DARKER than the input image.
  - The effect of BRIGHT details in the input image that are smaller in area than the structuring element is reduced, with the degree of reduction being determined by the grayscale values surrounding by the bright detail and by shape and amplitude values of the structuring element itself.
- Similar to binary image grayscale erosion and dilation are DUALS with respect to function complementation and reflection.

Opening and Closing

- Opening

rolling a disk

erosion dilation

$$f \circ b = (f \ominus b) \oplus b.$$

- Closing

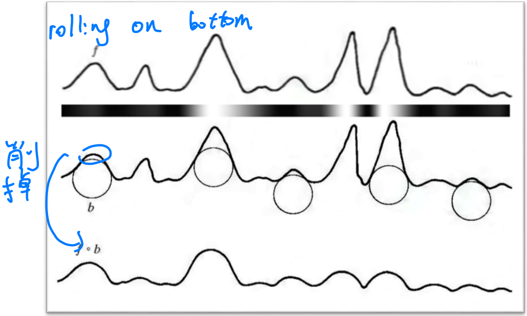
$$f \bullet b = (f \oplus b) \ominus b.$$

- In the opening of a gray-scale image, we remove small light details, while relatively undisturbed overall gray levels and larger bright features
- In the closing of a gray-scale image, we remove small dark details, while relatively undisturbed overall gray levels and larger dark features

back = background
white = objects

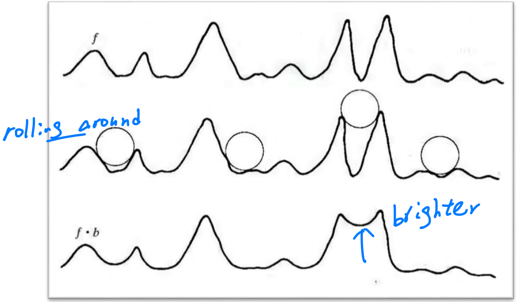
Opening

- Opening a gray scale picture describable as pushing object B under the scan-line graph, while traversing the graph according the curvature of B

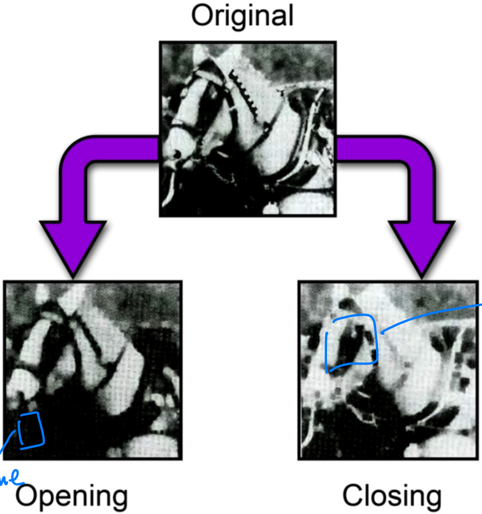


Closing

- Closing a gray scale picture is describable as pushing object B on top of the scan-line graph, while traversing the graph according the curvature of B
- The peaks are usually remains in their original form



Opening and Closing examples



black details gone!

bright details gone

size not change

Applications of Gray-Scale Morphology I

x bright

x black

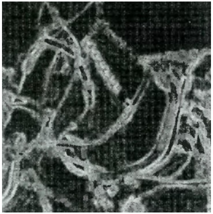
- Morphological smoothing
  - Perform opening followed by a closing
  - The net result of these two operations is to remove or attenuate both bright and dark artifacts or noise

Applications of Gray-Scale Morphology II

- Morphological gradient
  - Dilation and erosion are used to compute the morphological gradient of an image, denoted g:

$$g = (f \oplus b) - (f \ominus b)$$

  - It uses to highlights sharp gray-level transitions in the input image.
  - Obtained using symmetrical structuring elements tend to depend less on edge directionality.



Reading Assignment

- Chapter 9 of the textbook